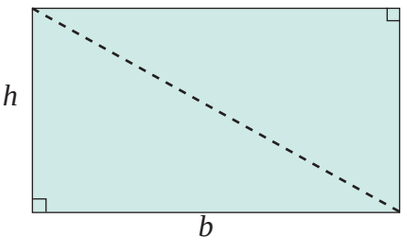
When a rectangular sheet is cut along its diagonal, two right angled triangles are obtained.

Area of two right angled triangles = Area of the rectangle

2 x Area of a right angled triangle = l x b

Area of the right angled triangle = $\frac{1}{2}$ (l x b) *sq. units.*

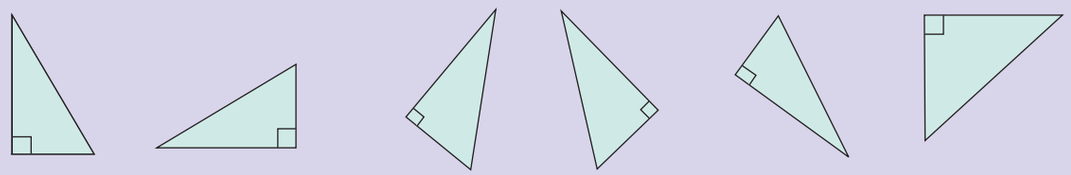


The length and breadth of the rectangle are respectively the base (b) and height (h) of the right angled triangle.

Hence, area of the right angled triangle = $\frac{1}{2}$ (b x h) *sq. units.*

Activity

Mark the base and height of the following right angled triangles.



Example 11 Find the area of a right angled triangle whose base is 18 *cm* and height is 12 *cm*.

Solution

Base, b = 18 *cm*

Height, h = 12 *cm*

Area, A = $\frac{1}{2}$ (b x h) *sq. units*

= $\frac{1}{2}$ (18 x 12)

= 108 *sq. cm*. (or) 108 cm^2

E7KS3Y

Try these

Draw the following in a graph sheet.

i) Two different rectangles whose areas are 16 cm^2 each.

ii) A shape with perimeter 14 *cm* and area 12 *sq. cm*.

iii) A shape with area 36 *sq. cm*.

iv) Form different shapes using 4 *unit* squares and find their perimeter and area. (Sides of the squares must fit exactly)

3.4 Perimeter and Area of Combined Shapes

A ***Combined shape*** is the combination of several closed shapes. The perimeter is calculated by adding all the outer sides (boundaries) of the combined shape. The area is calculated by adding all the areas of closed shapes from which the combined shape is formed.

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